

Lesson 17 — True or False? (+ Debug)

Matches LEGO Teacher Portal lesson: "Password Problem".

Objective (I can...): I can tell whether a condition is true or false and predict what my program will do.

CS concept: Boolean conditions (true/false) · **Builds on:** Lesson 16 (if/then).

AI Connection (2 min): Under the hood, an AI's "is this a cat?" answer becomes a true/false decision after it checks its confidence. *First-principles question:* "How sure should it be before it answers 'true'?"

Materials: LEGO CS & AI kit + device per pair, true/false cards. **Model reuse:** Conditionals model from Lesson 16 — code only.

60-minute flow - Warm-up (8): True/false flash — coach reads statements ("The hub is on"); pairs flip a card. - **Concept + demo (10):** Show the same if/then; before each run ask "is the condition true right now?" then run to confirm. - **Build/Code (30):** Pairs run their if/then in a true state and a false state, predicting first each time. **Debug mini-task (3 min):** give a program whose condition is set backwards (acts when it shouldn't) — pairs diagnose and fix. *Swap at 15 min.* - **Challenge (7):** Add a light that shows "condition true" so the state is visible. - **Share + pack-up (5):** Pairs predict, then demonstrate, both a true and false outcome. Model stays built.

Pair roles: Driver runs and codes, Navigator predicts true/false before each run. Swap at 15 min. (*Trio: third leads the debug.*)

Differentiation: *Support* — a checklist for reading the condition. *Extension / fast-finisher* — use a sensor threshold as the condition. *Reteach* — watch the live sensor value and say "true/false" as the object moves.

Exit check: Pair correctly predicts and shows the action for both true and false states. **At-home (optional):** Write 5 statements about your day; mark each true or false. ~10 min.
